

- Qu. 1.** a) Find the Cartesian and cylindrical polar coordinates of the point P which has spherical coordinates $(3, \pi/4, \pi/3)$.
- b) Find the spherical polar coordinates of the point Q which has Cartesian coordinates $(1, 1, 1)$.
- c) The surface, S , of a sphere of radius R , centre the origin, can be expressed in spherical polars coordinates as the set $\{(\rho, \varphi, \theta) : \rho = R, 0 \leq \varphi \leq \pi, 0 \leq \theta < 2\pi\}$. Find expressions for S in terms of Cartesian and cylindrical polar coordinates.
- d) The surface, C , of a cylinder of radius R with axis lying along the z -axis can be expressed in terms of cylindrical polar coordinates as the set $\{(r, \theta, z) : r = R, 0 \leq \theta < 2\pi, -\infty < z < \infty\}$. Find expressions for C in terms of Cartesian and spherical polar coordinates.
- e) Let K be the surface of an infinite cone with circular cross section, vertex at the origin and axis lying along the positive z -axis. If the angle between the z -axis and the surface of the cone is α , find expressions for K in terms of Cartesian, spherical and cylindrical polar coordinates.

Solution. a) $P = (\rho, \varphi, \theta) = (3, \pi/4, \pi/3)$, so

$$\begin{aligned} x &= \rho \sin \varphi \cos \theta = 3/2\sqrt{2} & y &= \rho \sin \varphi \sin \theta = 3\sqrt{3}/2\sqrt{2} \\ z &= \rho \cos \varphi = 3/\sqrt{2} & r &= \rho \sin \varphi = 3/\sqrt{2} & \theta &= \pi/3 \end{aligned}$$

Therefore, in Cartesian coordinates $P = (\frac{3}{2\sqrt{2}}, \frac{3\sqrt{3}}{2\sqrt{2}}, \frac{3}{\sqrt{2}})$ and in cylindrical polars $P = (\frac{3}{\sqrt{2}}, \frac{\pi}{3}, \frac{3}{\sqrt{2}})$. (Note, that these answers are precise, an answer in terms of decimals will always be an approximation and therefore not correct).

- b) Since $\rho \geq 0$ and $\rho^2 = x^2 + y^2 + z^2$, $\rho = \sqrt{3}$. $\tan \theta = \frac{y}{x} = 1$ so that $\theta = \pi/4$ or $5\pi/4$, but $0 \leq \theta \leq \pi/2$ since x and y are both positive, so $\theta = \pi/4$. Finally $\tan \varphi = \frac{r}{z}$, where $r = \sqrt{x^2 + y^2} = \sqrt{2}$, so $\tan \varphi = \sqrt{2}$. Since z is positive, $0 \leq \varphi \leq \pi/2$ and, therefore, $\varphi = \tan^{-1} \sqrt{2}$. Thus in spherical polars, $Q = (\sqrt{3}, \pi/4, \tan^{-1} \sqrt{2})$. (Since $\tan^{-1} \sqrt{2}$ cannot be expressed neatly in terms of π , a decimal solution here is perhaps more acceptable, but $\tan^{-1} \sqrt{2}$ is the correct answer).
- c) The equation of a sphere, centre $(0, 0, 0)$, radius R , in Cartesian coordinates is $x^2 + y^2 + z^2 = R^2$, so in terms of Cartesian coordinates $S = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$. In terms of cylindrical polars, θ is free to range over $[0, 2\pi)$ and we know that $\rho^2 = r^2 + z^2$ so $S = \{(r, \theta, z) : 0 \leq \theta < 2\pi, r^2 + z^2 = R^2\}$.
- d) We have $R^2 = r^2 = x^2 + y^2$, so in terms of Cartesian coordinates $C = \{(x, y, z) : x^2 + y^2 = R^2, -\infty < z < \infty\}$. For spherical polars, $r = R = \rho \sin \varphi$ so that $C = \{(\rho, \varphi, \theta) : \rho = \frac{R}{\sin \varphi}, 0 \leq \varphi \leq \pi, 0 \leq \theta < 2\pi\}$.

- e) In terms of spherical polars, $K = \{(\rho, \varphi, \theta) : 0 \leq \rho < \infty, \varphi = \alpha, 0 \leq \theta < 2\pi\}$. By simple trigonometry, $\tan \alpha = r/z = (\sqrt{x^2 + y^2})/z$. In terms of Cartesian coodinates then, $K = \{(x, y, z) : x^2 + y^2 = z \tan \alpha, 0 \leq z \leq \infty\}$ and, in terms of cylindrical polars, $K = \{(r, \theta, z) : r = z \tan \alpha, 0 \leq z \leq \infty, 0 \leq \theta < 2\pi\}$.

Qu. 2. Verify that:

- the element of area $dA = dx dy$ is equal to $r dr d\theta$ in polar co-ordinates;
- the element of volume $dV = dx dy dz$ is equal to $r dr d\theta dz$ in cylindrical polar co-ordinates;
- the element of volume $dV = dx dy dz$ is equal to $\rho^2 \sin \varphi d\rho d\varphi d\theta$ in spherical polar co-ordinates.

Solution. (1) In two dimensional polar co-ordinates, $x = r \cos \theta$ and $y = r \sin \theta$. The Jacobian

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

So that $dA = \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| dr d\theta = |r| dr d\theta = r dr d\theta$.

(2) In three dimensional cylindrical polar co-ordinates, $x = r \cos \theta$, $y = r \sin \theta$ and $z = z$. The Jacobian

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} x_r & x_\theta & x_z \\ y_r & y_\theta & y_z \\ z_r & z_\theta & z_z \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r$$

So that $dV = \left| \frac{\partial(x, y, z)}{\partial(r, \theta, z)} \right| dr d\theta dz = r dr d\theta dz$.

(3) In three dimensional spherical polar co-ordinates $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$ and $z = \rho \cos \varphi$. The Jacobian

$$\frac{\partial(x, y, z)}{\partial(\rho, \varphi, \theta)} = \begin{vmatrix} x_\rho & x_\varphi & x_\theta \\ y_\rho & y_\varphi & y_\theta \\ z_\rho & z_\varphi & z_\theta \end{vmatrix} = \begin{vmatrix} \sin \varphi \cos \theta & \rho \cos \varphi \cos \theta & -\rho \sin \varphi \sin \theta \\ \sin \varphi \sin \theta & \rho \cos \varphi \sin \theta & \rho \sin \varphi \cos \theta \\ \cos \varphi & -\rho \sin \varphi & 0 \end{vmatrix} = \rho^2 \sin \varphi.$$

So that $dV = \left| \frac{\partial(x, y, z)}{\partial(\rho, \varphi, \theta)} \right| d\rho d\varphi d\theta = |\rho^2 \sin \varphi| d\rho d\varphi d\theta = \rho^2 \sin \varphi d\rho d\varphi d\theta$. Note that for $\varphi \in [0, \pi]$, $\sin \varphi \geq 0$.

Qu. 3. Evaluate the following double integral as a repeated integral:

$\iint_R (x^2 + y^2) dA$ where R is the sector $\{(r, \theta) : 1 < r < 2, \pi/3 < \theta < \pi/2\}$ [Hint: use polar coordinates]

Solution.

Integrating with respect to r first and recalling that the element of area for polar coordinates is $rdrd\theta$, we have

$$\begin{aligned}\iint_R (x^2 + y^2) dA &= \int_{\theta=\pi/3}^{\theta=\pi/2} \left(\int_{r=1}^{r=2} r^2 r dr \right) d\theta = \int_{\theta=\pi/3}^{\theta=\pi/2} \left[\frac{r^4}{4} \right]_{r=1}^{r=2} d\theta \\ &= \int_{\theta=\pi/3}^{\theta=\pi/2} \frac{15}{4} d\theta = \left[\frac{15}{4} \theta \right]_{\theta=\pi/3}^{\theta=\pi/2} = \frac{\pi}{6} \frac{15}{4} = \frac{5\pi}{8}\end{aligned}$$

Qu. 4. a) The domain bounded by the surface of a paraboloid $z = 2 - x^2 - y^2$ and that of a cone $z^2 = x^2 + y^2$ is given by $D = \{x, y, z : x^2 + y^2 \leq 1, \sqrt{x^2 + y^2} \leq z \leq 2 - x^2 - y^2\}$. Find its volume using the appropriate coordinate system.

Solution. a) The appropriate coordinates one might wish to use are cylindrical coordinates (r, θ, z) , where the element of volume has the form $dx dy dz = r dr d\theta dz$.

Then the volume is given by the integral

$$V = \int_{r=0}^1 r dr \int_{\theta=0}^{2\pi} d\theta \int_{z=r}^{2-r^2} dz = 2\pi \int_0^1 r[(2 - r^2) - r] dr = 2\pi \left(1 - \frac{1}{4} - \frac{1}{3}\right) = \frac{5\pi}{6}.$$

b) After changing to polar coordinates and having in mind that $\tan^{-1}(\frac{y}{x}) = \theta$, $dx dy = r dr d\theta$ and the boundaries $y = 0$, $y = x$ and $x = 1$ are in polar coordinates described as $\theta = 0$, $\theta = \pi/4$ and $r = 1/\cos \theta$ (or alternatively $\theta = \cos^{-1}(1/r)$), respectively, we have

$$\begin{aligned}\iint_{\substack{0 \leq x \leq 1 \\ 0 \leq y \leq x}} \tan^{-1}\left(\frac{y}{x}\right) dx dy &= \int_0^{\pi/4} \theta d\theta \int_0^{1/\cos \theta} r dr \\ &= \frac{1}{2} \int_0^{\pi/4} \frac{\theta d\theta}{\cos^2 \theta} = \frac{1}{2} \int_0^{\pi/4} \theta d(\tan \theta) = \left[\frac{\theta}{2} \tan \theta \right]_{\theta=0}^{\pi/4} - \frac{1}{2} \int_0^{\pi/4} \tan \theta d\theta \\ &= \frac{\pi}{8} + \frac{1}{2} \int_0^{\pi/4} \frac{d \cos \theta}{\cos \theta} = \frac{\pi}{8} + \frac{1}{2} [\ln |\cos \theta|]_{\theta=0}^{\pi/4} = \frac{\pi}{8} - \frac{1}{4} \ln 2.\end{aligned}$$

Qu. 5. Evaluate each of the following triple integrals as a repeated integral using an appropriate coordinate systems:

a) $\iiint_R z e^{-(x^2+y^2+z^2)} dV$, where $R = \{(x, y, z) : -\infty < x, y < \infty, 0 \leq z < 1\}$.

b) $\iiint_S x^2 dV$, where S is the unit ball $\{(\rho, \varphi, \theta) : \rho < 1\}$.

Solution. a) The integrand looks as though cylindrical polar coordinates are the sensible choice. (Spherical coordinates are also an option, though a tricky one, whereas in Cartesian coordinates things get hard!). In cylindrical polar, $R = \{(r, \theta, z) : 0 < r < \infty, 0 \leq \theta < 2\pi, 0 \leq z \leq 1\}$ so, integrating with respect to θ first fixing r and z , then integrating with respect to r fixing z , and recalling that the element of volume $dV = r dr d\theta dz$, we have

$$\begin{aligned} I &= \iiint_R z e^{-(x^2+y^2+z^2)} dV = \iiint_R z e^{-z^2} e^{-(x^2+y^2)} dV \\ &= \int_{z=0}^{z=1} \int_{r=0}^{r=\infty} \int_{\theta=0}^{\theta=2\pi} z e^{-z^2} e^{-r^2} r d\theta dr dz \\ &= \int_{z=0}^{z=1} \int_{r=0}^{\infty} \left[\theta z e^{-z^2} r e^{-r^2} \right]_{\theta=0}^{\theta=2\pi} dr dz = 2\pi \int_{z=0}^{z=1} \int_{r=0}^{\infty} z e^{-z^2} r e^{-r^2} dr dz \\ &= 2\pi \int_{z=0}^{z=1} \left[-\frac{1}{2} z e^{-z^2} e^{-r^2} \right]_{r=0}^{r=\infty} dz \\ &= 2\pi \int_{z=0}^{z=1} \frac{1}{2} z e^{-z^2} dz = \pi \left[-\frac{1}{2} e^{-z^2} \right]_{z=0}^{z=1} = \frac{\pi}{2} (1 - e^{-1}) \end{aligned}$$

b) Clearly the question is asking us to use spherical polars. Integrating with respect to ρ , then φ , then θ , recalling that the element of volume is $dV = \rho^2 \sin \varphi d\rho d\varphi d\theta$ we have

$$\begin{aligned} \iiint_S x^2 dV &= \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi} \int_{\rho=0}^{\rho=1} (\rho \sin \varphi \cos \theta)^2 \rho^2 \sin \varphi d\rho d\varphi d\theta \\ &= \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi} \int_{\rho=0}^{\rho=1} \rho^4 \sin^3 \varphi \cos^2 \theta d\rho d\varphi d\theta \\ &= \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi} \int_{\rho=0}^{\rho=1} \rho^4 (\sin \varphi (1 - \cos^2 \varphi)) \left(\frac{1}{2} (1 + \cos 2\theta) \right) d\rho d\varphi d\theta \\ &= \frac{1}{2} \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi} \left[\frac{1}{5} \rho^5 (\sin \varphi (1 - \cos^2 \varphi)) (1 + \cos 2\theta) \right]_{\rho=0}^{\rho=1} d\varphi d\theta \\ &= \frac{1}{10} \int_{\theta=0}^{\theta=2\pi} \int_{\varphi=0}^{\varphi=\pi} (\sin \varphi (1 - \cos^2 \varphi)) (1 + \cos 2\theta) d\varphi d\theta \\ &= \frac{1}{10} \int_{\theta=0}^{\theta=2\pi} \left[(-\cos \varphi + \frac{1}{3} \cos^3 \varphi) (1 + \cos 2\theta) \right]_{\varphi=0}^{\varphi=\pi} d\theta \\ &= \frac{1}{10} \int_{\theta=0}^{\theta=2\pi} \frac{4}{3} (1 + \cos 2\theta) d\theta = \frac{2}{15} \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\theta=0}^{\theta=2\pi} = \frac{4\pi}{15} \end{aligned}$$

Qu. 6. Find the volume between the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 1$ that lies in the first octant (i.e. $x > 0, y > 0, z > 0$) using cylindrical coordinates.

Solution. The circumference at which the cone intersects the sphere when both equations are satisfied $x^2 + y^2 = z^2$ and $x^2 + y^2 + z^2 = 1$. Then, we have $r = z = 1/\sqrt{2}$. In cylindrical polars, the equations of the cone and the sphere are $z = r$ and $z = \sqrt{1 - r^2}$. Then, remembering that $dV = r dr d\theta dz$, we have

$$\begin{aligned} V &= \iiint_V dV = \int_0^{\pi/2} d\theta \int_0^{1/\sqrt{2}} r dr \int_r^{\sqrt{1-r^2}} dz = \frac{\pi}{2} \int_0^{1/\sqrt{2}} r [z]_r^{\sqrt{1-r^2}} dz = \frac{\pi}{2} \int_0^{1/\sqrt{2}} (r\sqrt{1-r^2} - r^2) dr \\ &= -\frac{\pi}{4} \int_0^{1/\sqrt{2}} \sqrt{1-r^2} d(1-r^2) - \int_0^{1/\sqrt{2}} r^2 dr = \frac{\pi}{6} \left(1 - \frac{1}{\sqrt{2}}\right) = \frac{\pi}{12}(2 - \sqrt{2}). \end{aligned}$$

Alternatively, integration can be carried out by integrating with respect to r first:

$$V = \underbrace{\int_0^{\pi/2} d\theta \int_0^{1/\sqrt{2}} dz \int_0^z r dr}_{=V_1} + \underbrace{\int_0^{\pi/2} d\theta \int_{1/\sqrt{2}}^1 dz \int_0^{\sqrt{1-z^2}} r dr}_{=V_2}.$$

Here V_1 is the volume of the cone contained between the coordinate plane $z = 0$ and the plane $z = 1/\sqrt{2}$ (and lying in the first quadrant). Volume V_2 is the volume of the spherical segment located above the plane $z = 1/\sqrt{2}$ (lying in the first quadrant). For these volumes one has:

$$\begin{aligned} V_1 &= \frac{\pi}{2} \int_0^{1/\sqrt{2}} \left[\frac{r^2}{2} \right]_0^z dz = \frac{\pi}{4} \int_0^{1/\sqrt{2}} z^2 dz = \frac{\pi\sqrt{2}}{48}. \\ V_2 &= \frac{\pi}{2} \int_{1/\sqrt{2}}^1 \left[\frac{r^2}{2} \right]_0^{\sqrt{1-z^2}} dz = \frac{\pi}{4} \int_{1/\sqrt{2}}^1 (1 - z^2) dz = \frac{\pi}{48}(8 - 5\sqrt{2}). \end{aligned}$$

Finally,

$$V = V_1 + V_2 = \frac{\pi\sqrt{2}}{48} + \frac{\pi}{48}(8 - 5\sqrt{2}) = \frac{\pi}{12}(2 - \sqrt{2}).$$